HK IMO Training 2021-2022
Problem Set 5 Suggested Solutions

Problem 17. Let p be an odd prime. Initially, we have two integers chosen from $S = \{1, 2, \ldots, p-1\}$. In each step, if the current integers we have are a and b, we can replace a by the unique integer $c \in S$ such that $bc \equiv a \pmod{p}$, or replace b by the unique integer d such that $ad \equiv b \pmod{p}$. Prove that we can make the two integers the same after finitely many steps.

Solution. Let g be a primitive root modulo p. If the current integers are g^x and g^y modulo p where $1 \leqslant x \leqslant y \leqslant p-1$, then the operation means we can replace g^x by g^{x-y} modulo p or replace g^y by g^{y-x} modulo p. Our goal is to make $x = y$.

We are done if $x = y$. So we may assume $x < y$. We replace g^y by g^{y-x} modulo p. Note that $1 \leqslant y - x < y < p-1$, so the new exponents x and $y - x$ are positive and have a smaller sum than the old sum $x + y$. The same cannot happen indefinitely. This means we must eventually obtain two integers with the same exponents. So we are done. $\square$

Problem 18 (Rioplatense Mathematical Olympiad 2019 Problem 2)**.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\big(f(x)^2 + f(y^2)\big) = (x - y)f(x - f(y)). \tag{1}$$

Solution. The solutions are $f(x) = 0$ for all $x \in \mathbb{R}$ and $f(x) = -x$ for all $x \in \mathbb{R}$. It is easy to check that both are solutions.

Replacing y by $-y$ in (1), and comparing with (1), we obtain

$$(x - y)f(x - f(y)) = (x + y)f(x - f(-y)). \tag{2}$$

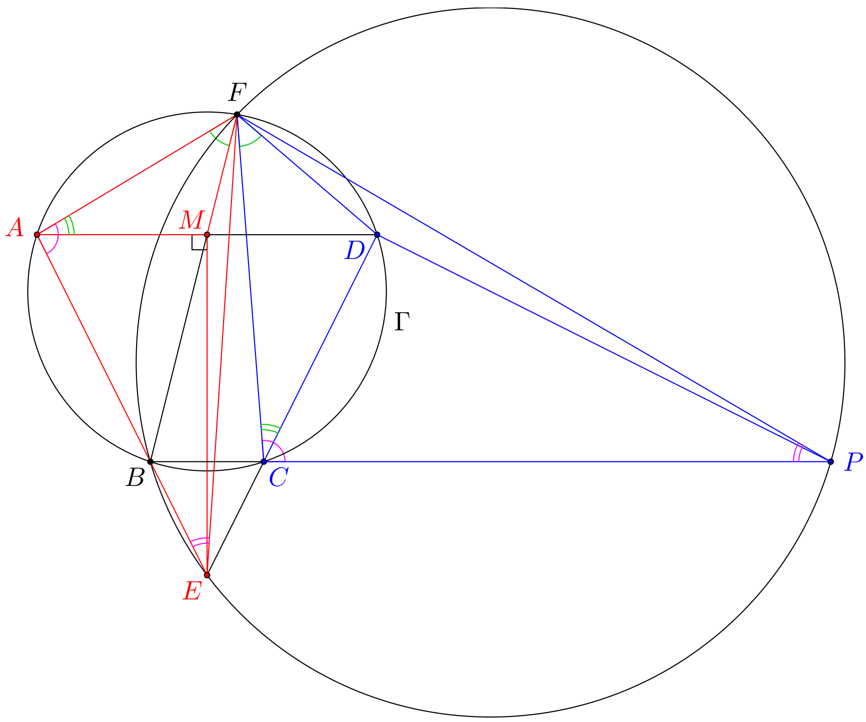
If $f(t) = 0$ for some $t \neq 0$, by putting $x = y = -t$ in (2), we get $f(-t) = 0$. Thus, putting $y = t$ in (2), we find that

$$(x - t)f(x) = (x + t)f(x).$$

As $x - t \neq x + t$, this implies $f(x) = 0$ for all $x \in \mathbb{R}$.

If $f(t) = 0$ only if $t = 0$, we put $x = y \neq 0$ in (2) to obtain $f(x - f(-x)) = 0$ for $x \neq 0$. This implies $x - f(-x) = 0$ for all $x \neq 0$, and hence $f(x) = -x$ for all $x \neq 0$. As $f(t) = 0$ for some $t = x - f(-x)$, we must have $f(0) = 0$. So $f(x) = -x$ for all $x \in \mathbb{R}$. $\square$

Problem 19 (239 Open MO 2012 Senior Problem 5)**.** Let $ABCD$ be a trapezoid inscribed in a circle Γ, where $AD /\!/ BC$. The rays AB and DC intersect at E. Let F be the point on Γ such that BF bisects AD. The circumcircle of $\triangle BEF$ intersects the ray BC again at P. Prove that $\angle EDP = 90°$.



Solution. Let M be the midpoint of AD. Note that $\angle AFM = \angle CFD$ since they are opposite to the equal chords AB and CD. Also, we have $\angle MAF = \angle DCF$. Therefore, $\triangle MAF \sim \triangle DCF$. Next, we have $\angle EAF = \angle PCF$ and $\angle FEA = \angle FPC$. This implies $\triangle EAF \sim \triangle PCF$. Combining the two pairs of similar triangles, we see that $EMAF \sim PDCF$. Therefore, $\angle CDP = \angle AME = 90^\circ$, where the last equality holds as $\triangle EAD$ is isosceles. $\square$

Problem 20 (239 Open MO 2017 Senior Problem 7)**.** Let n and k be positive integers. Two players X and Y play a game on a $2n \times 2n$ chessboard as follows. Firstly, X chooses one of the cells secretly without letting Y know which cell it is. We call this the position of X. In each round, the following things happen in order.

- X moves to an adjacent cell secretly. Note that an adjacent cell to C means a cell sharing a common edge with C which is different from C.
- Y chooses k cells. If one of the cells chosen by Y is the position of X, then Y wins the game.

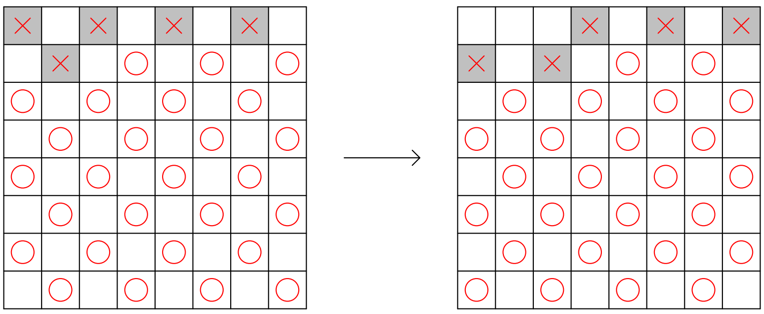
Find the smallest possible value of k such that Y can always win the game after a finite number of rounds.

Solution. The answer is $n+1$.

Firstly, we show that Y has a winning strategy when $k = n+1$. We use the usual chessboard colouring such that the top left corner cell is black. Note that we may assume Y always knows the colour of the current cell of X. Indeed, Y can first apply the strategy below by assuming X's initial cell is black. If it turns out that Y cannot win after applying the strategy, that means X's initial cell must be white. Then Y can apply the same strategy to win the game. We label all the cells by $1, 2, \ldots, 4n^2$ from top to bottom and from left to right (so the cells in the first row are $1, 2, \ldots, 2n$).

Let S_j be the labels of all possible positions of X after X's jth turn (under the assumption of the colour of X's cell). Initially, assume S_1 contains the labels of all black cells. We now give a strategy of Y such that S_j eventually becomes the empty set. Indeed, in each round, Y just needs to choose the $n+1$ cells in S_j with the smallest labels.

Let $\min S_j = m$. We claim that $\min S_{j+1} = m' \geqslant m+1$. Since $m' \in S_{j+1}$, one of $m'-2n$, $m'-1$, $m'+1$ or $m'+2n$ must belong to S_j since these are the only possible labels of the adjacent cells to cell m' (actually, some may not be adjacent to cell m' if cell m' lies on the boundary). Also, this cell must survive after Y has chosen $n+1$ cells in the jth round, or otherwise Y has already won the game. Therefore, there are at least $n+1$ cells having the same colour as cell m whose labels are less than $m'+2n$. But then it is not hard to see that there are at most n cells of the same colour among cells $m'+1, m'+2, \ldots, m'+2n-1$. As $m = \min S_j$, this implies $m < m'+1$. Also, since cells m and m' have different colours, we have $m \neq m'$. Thus, $m \leqslant m'-1$, which proves our claim. Therefore, $\min S_j$ is strictly increasing, and eventually it must become the empty set since the label is bounded above. So Y has a winning strategy when $k = n+1$.



Secondly, we show that Y does not have a winning strategy when $k = n$, which clearly proves $n+1$ is the best possible. Again, Y may assume X's initial cell is black, and we define S_j as before. It suffices to prove that whenever $|S_j| \geqslant n^2+n$, then $|S_{j+1}| \geqslant n^2+n$ (since $|S_1| = 2n^2 \geqslant n^2+n$ and $n^2+n > n$, this shows Y can never guarantee to find X's cell by choosing only n cells each time). In other words, we want to show that for any set A of n^2 cells of the same colour (say, black), there are at least n^2+n cells which are adjacent to at least one of these cells. The case $n=1$ is trivial. We assume $n \geqslant 2$ in the following.

Let $a_1, a_2, \ldots, a_{2n}$ be the number of cells in A lying in columns $1, 2, \ldots, 2n$ respectively. Let $b_1, b_2, \ldots, b_{2n}$ be the number of cells in columns $1, 2, \ldots, 2n$ which are adjacent to at least one cell in A. We make the following observations.

(i) $0 \leqslant a_k, b_k \leqslant n$ for any k

(ii) $b_k \geqslant \max\{a_{k-1}, a_k, a_{k+1}\}$ for any k (with $a_0 = a_{2n+1} = 0$)

(iii) $b_k + b_{k+1} \geqslant a_k + a_{k+1} + 1$ for any k unless $(a_k, a_{k+1}) = (0,0), (n,n)$

(iv) if $a_i < a_j$ for some i, j (say $i < j$), then $(b_i + b_{i+1} + \cdots + b_{j-1}) - (a_i + a_{i+1} + \cdots + a_{j-1}) \geqslant a_j - a_i$

(i) is obvious. For (ii), $b_k \geqslant a_{k-1}$ because the cells in column $k-1$ can be shifted rightwards (similarly for $b_k \geqslant a_{k+1}$. Also, $b_k \geqslant a_k$ because the cells in column k can be shifted upwards or downwards (depending on the colour). In particular, $b_k = a_k$ only if the black cells of A in column k are all at the top or at the bottom (with one cell on the boundary of the chessboard).

For (iii), suppose $(a_k, a_{k+1}) \neq (0,0), (n,n)$. Note that we must have $b_k + b_{k+1} \geqslant a_k + a_{k+1}$ by (ii). If equality holds, WLOG we may assume the black cells of A in column k are all at the top, and the black cells of A in column $k+1$ are all at the bottom. Since the top cell in column k can be shifted rightwards and the bottom cell in column $k+1$ can be shifted leftwards, there must be a new adjacent cell which is not yet counted. Thus, (iii) holds.

(iv) follows from (ii) since

$$(b_i + b_{i+1} + \cdots + b_{j-1}) - (a_i + a_{i+1} + \cdots + a_{j-1}) \geqslant (a_{i+1} + a_{i+2} + \cdots + a_j) - (a_i + a_{i+1} + \cdots + a_{j-1}) = a_j - a_i.$$

Suppose on the contrary that $(b_1 + b_2 + \cdots + b_{2n}) - (a_1 + a_2 + \cdots + a_{2n}) \leqslant n - 1$. Partitioning the columns into pairs $(1,2), (3,4), \ldots, (2n-1, 2n)$, we see by (iii) that there must be a pair such that $a_{2i} = a_{2i+1} = 0$ or $a_{2i} = a_{2i+1} = n$. By (iv), exactly one of these may happen (otherwise, there are at least n more new cells between the columns with 0 and n cells in A).

Case 1: $a_{2i+1} = a_{2i+2} = 0$ for some i

Let $i \leqslant j$ be the smallest and largest indices such that $a_{2i+1} = a_{2i+2} = a_{2j+1} = a_{2j+2} = 0$. Define

$$\begin{aligned}
x &= (b_1 + b_2 + \cdots + b_{2i+1}) - (a_1 + a_2 + \cdots + a_{2i+1}), \\
y &= (b_{2i+2} + b_{2i+3} + \cdots + b_{2j+1}) - (a_{2i+2} + a_{2i+3} + \cdots + a_{2j+1}), \\
z &= (b_{2j+2} + b_{2j+3} + \cdots + b_{2n}) - (a_{2j+2} + a_{2j+3} + \cdots + a_{2n}).
\end{aligned}$$

(If some parts do not exist, we define the corresponding difference to be 0.) Note that $x + y + z \leqslant n - 1$.

By (iii), there is a new cell in each pair of columns with labels $(1,2), (3,4), \ldots, (2i-3, 2i-2)$. So there are at most $x - (i-1)$ new cells in columns $2i-1, 2i, 2i+1$. By (iv), we have $a_{2i-1}, a_{2i} \leqslant x - (i-1)$ (note that we must have $x \geqslant i$). Similarly, we have $a_{2i-3}, a_{2i-2} \leqslant x - (i-2)$, etc. We wish to obtain an upper bound for $a_1 + a_2 + \cdots + a_{2n}$. So we may assume $a_1 = a_2 = x, a_3 = a_4 = x - 1, \ldots, a_{2i-1} = a_{2i} = x - i + 1$. Similarly, assume $a_{2n} = a_{2n-1} = z, a_{2n-2} = a_{2n-3} = z - 1, \ldots, a_{2j+4} = a_{2j+3} = z - n + j + 2$. Indeed, we can also 'combine' these two parts together (in the sense that $j = n - 1$ so that the right part does not exist) because

$$\begin{aligned}
&2(x + (x-1) + \cdots + (x-i+1)) + 2(z + (z-1) + \cdots + (z-n+j+2)) \\
&\leqslant 2((x+z) + (x+z-1) + \cdots + (x+z-i+1) + (x+z-i) + (x+z-i-1) + \cdots + (x+z-n+j-i+2))
\end{aligned}$$

(as $x \geqslant i$). Next, if $\max\{a_{2i+3}, a_{2i+4}, \ldots, a_{2n-2}\} = a_\ell = M$, then there are $\geqslant M$ new cells in columns $a_{2i+2}, a_{2i+3}, \ldots, a_{\ell-1}$ and $\geqslant M$ new cells in columns $a_{\ell+1}, a_{\ell+2}, \ldots, a_{2n-1}$ by (iv). Thus, we have $2M \leqslant y$, so $a_m \leqslant \dfrac{y}{2}$ for $2i + 3 \leqslant m \leqslant 2n - 2$. Now, we have

$$\begin{aligned}
n^2 &= a_1 + a_2 + \cdots + a_{2n} \\
&\leqslant 2(x + (x-1) + \cdots + (x-i+1)) + \frac{y}{2}(2n - 2i - 4) \\
&= (2x - i + 1)i + y(n - i - 2) \\
&\leqslant (2(n - 1 - y) - i + 1)i + y(n - i - 2) \\
&= (n - 3i - 2)y + (2n - i - 1)i.
\end{aligned}$$

If $n \geqslant 3i + 2$, this implies $n^2 \leqslant (n - 3i - 2)(n - i - 1) + (2n - i - 1)i$ since $y \leqslant n - 1 - x \leqslant n - 1 - i$. This yields $2i^2 + 4i + 2 \geqslant (2i+3)n \geqslant (2i+3)(3i+2)$. This is impossible.

If $n < 3i + 2$, this implies $n^2 \leqslant (2n - i - 1)i$, i.e. $(n-i)^2 + i \leqslant 0$. This is again impossible.

Case 2: $a_{2i+1} = a_{2i+2} = n$ for some i

Let $i \leqslant j$ be the smallest and largest indices such that $a_{2i+1} = a_{2i+2} = a_{2j+1} = a_{2j+2} = n$. Define

$$\begin{aligned}
x &= (b_1 + b_2 + \cdots + b_{2i+1}) - (a_1 + a_2 + \cdots + a_{2i+1}), \\
y &= (b_{2i+2} + b_{2i+3} + \cdots + b_{2j+1}) - (a_{2i+2} + a_{2i+3} + \cdots + a_{2j+1}), \\
z &= (b_{2j+2} + b_{2j+3} + \cdots + b_{2n}) - (a_{2j+2} + a_{2j+3} + \cdots + a_{2n}).
\end{aligned}$$

Note that $x + y + z \leqslant n - 1$.

By (iii), there is a new cell in each pair of columns with labels $(1, 2), (3, 4), \ldots, (2i-3, 2i-2)$. So there are at most $x - (i-1)$ new cells in the columns $2i - 1$ and $2i$. By (iv), we have $a_{2i-1}, a_{2i} \geqslant n - (x - (i-1))$ since $a_{2i+1} = n$. Similarly, we have $a_{2i-3}, a_{2i-2} \geqslant n - (x - (i-2))$, etc. Since we wish to find a lower bound for $a_1 + a_2 + \cdots + a_{2n}$, we may assume $a_1 = a_2 = n - x$, $a_3 = a_4 = n - x + 1, \ldots, a_{2i-1} = a_{2i} = n - x + i - 1$. As in case 1, we can assume $j = n - 1$.

Next, let $a_{\ell_1} = m_1$ be the smallest among $a_{2i+3}, a_{2i+4}, \ldots, a_{2n-2}$, and let $a_{\ell_2} = m_2$ be the smallest among these numbers after a_{ℓ_1} is removed. By (iv), then there are at least $n - m_1$ new cells from column ℓ_1 to one end (column $2i + 2$ or $2n - 1$), and at least $n - m_2$ new cells from column ℓ_2 to the other end. Thus, we have $2n - m_1 - m_2 \leqslant y$, and hence $2m_2 \geqslant m_1 + m_2 \geqslant 2n - y$. WLOG assume one of $a_{2i+3}, a_{2i+4}, \ldots, a_{2n-2}$ is m_1 and assume the others are m_2.

- Firstly, we consider the case $i \leqslant n - 3$. Recall that $a_{2i+1} = a_{2i+2} = a_{2j+1} = a_{2j+2} = n$. So

$$\begin{aligned}
n^2 &= a_1 + a_2 + \cdots + a_{2n} \\
&\geqslant 2((n-x) + (n-x+1) + \cdots + (n-x+i-1)) + 4n + m_1 + m_2(2n-2i-5) \\
&= (2n-2x+i-1)i + 4n + (m_1+m_2) + m_2(2n-2i-6) \\
&\geqslant (2n - 2(n-1-y) + i - 1)i + 4n + (2n-y)(n-i-2) \\
&= (3i+2-n)y + (i+1)i + 2n(n-i).
\end{aligned}$$

  If $n \geqslant 3i + 2$, this implies $n^2 \geqslant (3i+2-n)(n-i-1) + (i+1)i + 2n(n-i)$ since $y \leqslant n - 1 - x \leqslant n - 1 - i$. This yields $2i^2 + 4i + 2 \geqslant (2i+3)n \geqslant (2i+3)(3i+2)$. This is impossible.

  If $n < 3i + 2$, this implies $n^2 \geqslant (i+1)i + 2n(n-i)$, i.e. $(n-i)^2 + i \leqslant 0$. This is again impossible.

- Secondly, we consider the case $i = n - 2$. Then columns $2i + 1, 2i + 2$ are adjacent to columns $2n - 1, 2n$, and so $y = 0$. Then we have

$$\begin{aligned}
n^2 &= a_1 + a_2 + \cdots + a_{2n} \\
&\geqslant 2((n-x) + (n-x+1) + \cdots + (n-x+n-3)) + 4n \\
&= (3n-2x-3)(n-2) + 4n \\
&\geqslant (3n - 2(n-1) - 3)(n-2) + 4n \\
&= n^2 + n + 2.
\end{aligned}$$

  This is impossible.

- Similarly, when $i = n - 1$, we have

$$\begin{aligned}
n^2 &= a_1 + a_2 + \cdots + a_{2n} \\
&\geqslant 2((n-x) + (n-x+1) + \cdots + (n-x+n-2)) + 2n \\
&= (3n-2x-2)(n-1) + 2n \\
&\geqslant (3n - 2(n-1) - 2)(n-1) + 2n \\
&= n^2 + n.
\end{aligned}$$

  This is impossible.

Thus, in both cases, we have deduced a contradiction. This completes the proof. $\square$

Remarks. Unluckily, this proof is entirely algebra (inequality + local refinement).